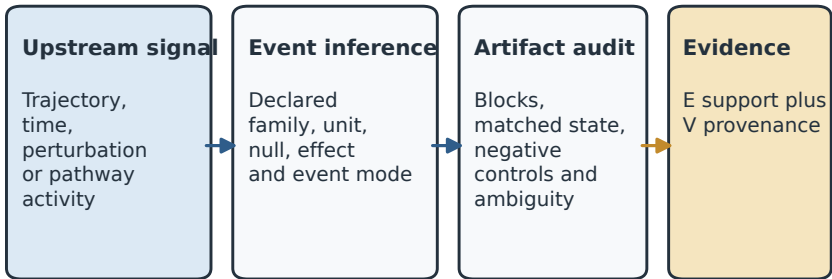


Problem definition and TED algorithm

A. From a signal to an interpretable event



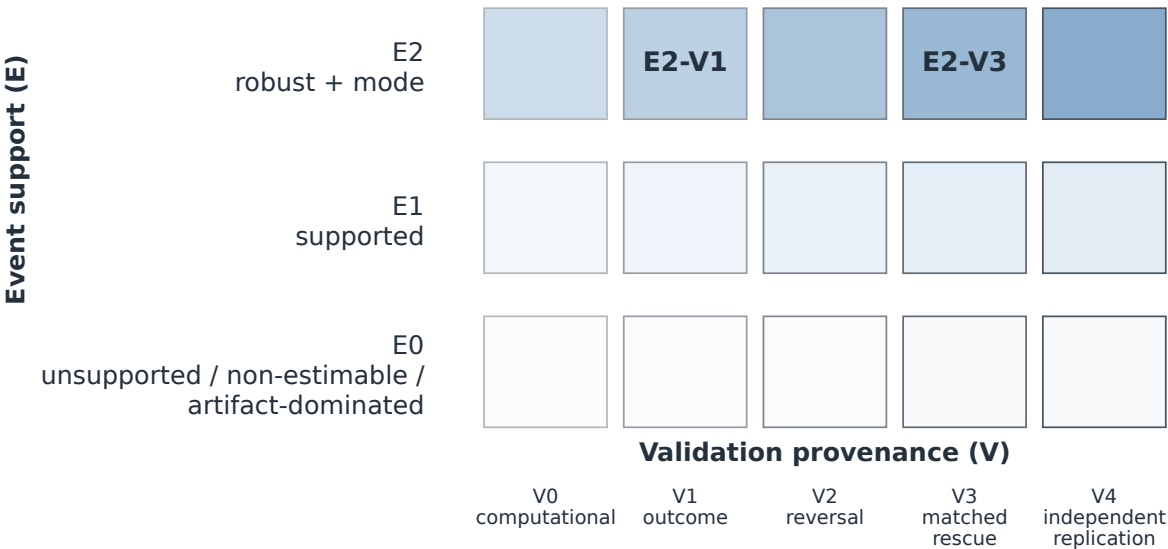
Detection, classification, artifact assessment and validation are separate targets

B. TED event-record fields

Identity	dataset; contrast; event family; analysis unit
Estimate	effect; direction; timing; test status; event q/missing reason
Robustness	block support; state overlap; control margin
Interpretation	event mode; ambiguity set; identifiability
Evidence	E code; V code; supported and unsupported interpretation

Missing optional evidence is recorded as unavailable, not as negative evidence.

C. Orthogonal evidence descriptor



D. Applicability and fail-closed behavior

Minimum requirements	Return E0 + reason / ambiguity
● Biological blocks	✗ E0_not_supported
● Declared event family	✗ E0_not_estimable
● Adequate state overlap	✗ E0_not_identifiable
● Plausible negative controls	✗ E0_artifact_dominated
● Interpretable analysis unit	✗ E0_missing_required_design

not_run: q is null + missing reason; valid run: q is numeric.
Large cell counts do not replace independent biological units.